PHYSICS I
WEEK 2



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IIT Madras

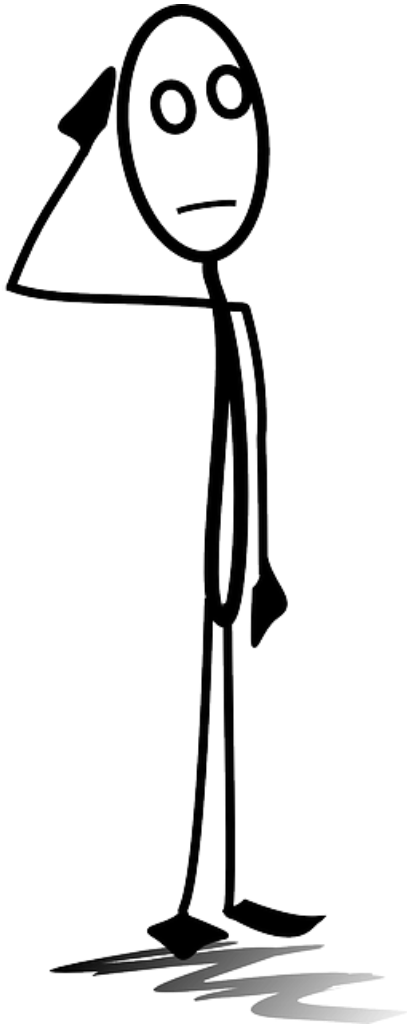
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WEEK 2: OUTLINE

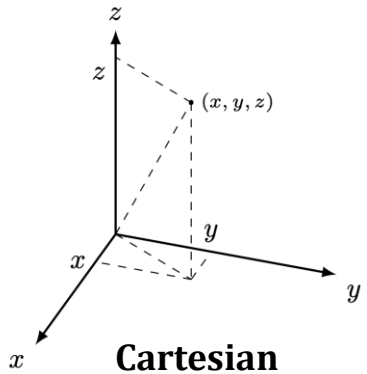
Newton's Laws

- First and second law, inertial, non-inertial frames: Taylor 1.4, 9.1
- Third law and conservation of momentum: Taylor 1.5
- Second law in polar co-ordinates: Taylor 1.6, 1.7

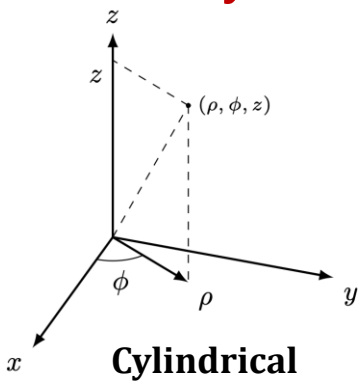


REVIEW

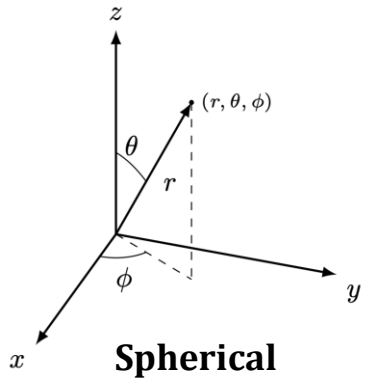
Co-ordinate systems



Cartesian

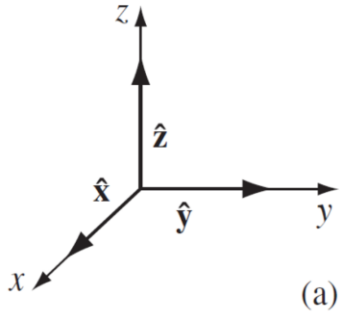


Cylindrical

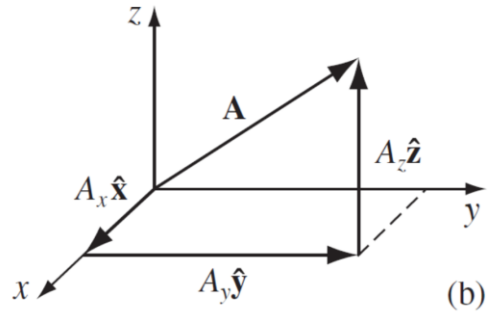


Spherical

Vectors



(a)

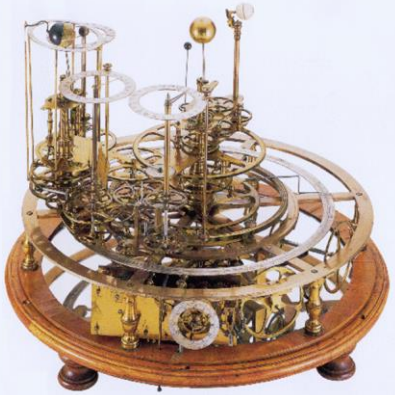


(b)

$$\mathbf{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}$$

Deterministic universe

Input:
Initial Conditions
$x(t_0)$ & $p(t_0)$



Laws of Physics

Output:
Final state
$x(t_f)$ & $p(t_f)$

REVISITING NEWTON'S LAWS: TIME

Time derivatives of vectors

$$\frac{d\mathbf{r}}{dt} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{r}}{\Delta t}$$

$$\Delta \mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$$

REVISITING NEWTON'S LAWS: TIME

Time derivatives of vectors

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$$\Delta \mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$$

Time dependent position vector

$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}},$$

$$\frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\hat{\mathbf{x}} + \frac{dy}{dt}\hat{\mathbf{y}} + \frac{dz}{dt}\hat{\mathbf{z}}.$$

Velocity vector

$$\mathbf{v} = v_x\hat{\mathbf{x}} + v_y\hat{\mathbf{y}} + v_z\hat{\mathbf{z}}$$

REVISITING NEWTON'S LAWS: TIME

Time derivatives of vectors

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Velocity vector

$$\mathbf{v} = v_x\hat{\mathbf{x}} + v_y\hat{\mathbf{y}} + v_z\hat{\mathbf{z}}$$

if $\mathbf{r}(t)$ and $\mathbf{s}(t)$ are two vectors that depend on t,

$$\frac{d}{dt}(\mathbf{r} + \mathbf{s}) = \frac{d\mathbf{r}}{dt} + \frac{d\mathbf{s}}{dt}$$

if $\mathbf{r}(t)$ is a vector and $f(t)$ is a scalar,

$$\frac{d}{dt}(f\mathbf{r}) = f\frac{d\mathbf{r}}{dt} + \frac{df}{dt}\mathbf{r}$$

REVISITING NEWTON'S LAWS: VECTORS

Sum: $\boldsymbol{F} = \boldsymbol{F_1} + \boldsymbol{F_2}$, $\boldsymbol{F} = (F_{1x} + F_{2x})\hat{\boldsymbol{x}} + (F_{1y} + F_{2y})\hat{\boldsymbol{y}} + (F_{1z} + F_{2z})\hat{\boldsymbol{z}}$

REVISITING NEWTON'S LAWS: VECTORS

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Multiplication with a scalar: $c\boldsymbol{F} = cF_x\hat{\boldsymbol{x}} + cF_y\hat{\boldsymbol{y}} + cF_z\hat{\boldsymbol{z}}$

REVISITING NEWTON'S LAWS: VECTORS

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Multiplication with a scalar: $c\boldsymbol{F} = cF_x\hat{\boldsymbol{x}} + cF_y\hat{\boldsymbol{y}} + cF_z\hat{\boldsymbol{z}}$

Dot product: Construct a scalar from two vectors

$$\boldsymbol{u} \cdot \boldsymbol{v} = |\boldsymbol{u}||\boldsymbol{v}|\cos\theta \quad \Rightarrow \quad \boldsymbol{u} \cdot \boldsymbol{u} = |\boldsymbol{u}|$$

REVISITING NEWTON'S LAWS: VECTORS

Sum: $\boldsymbol{F} = \boldsymbol{F_1} + \boldsymbol{F_2},\ \boldsymbol{F} = (F_{1x} + F_{2x})\hat{\boldsymbol{x}} + (F_{1y} + F_{2y})\hat{\boldsymbol{y}} + (F_{1z} + F_{2z})\hat{\boldsymbol{z}}$

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Cross product: Construct a vector from two vectors

$$|\boldsymbol{u} \times \boldsymbol{v}| = |\boldsymbol{u}||\boldsymbol{v}|\sin\theta = |\boldsymbol{w}| \quad \Rightarrow \quad \boldsymbol{u} \times \boldsymbol{u} = \boldsymbol{0}$$

REVISITING NEWTON'S LAWS: VECTORS

Sum: $\boldsymbol{F} = \boldsymbol{F_1} + \boldsymbol{F_2},\ \boldsymbol{F} = (F_{1x} + F_{2x})\hat{\boldsymbol{x}} + \left(F_{1y} + F_{2y}\right)\hat{\boldsymbol{y}} + (F_{1z} + F_{2z})\hat{\boldsymbol{z}}$

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$$|\boldsymbol{u} \times \boldsymbol{v}| = |\boldsymbol{u}||\boldsymbol{v}|\sin\theta = |\boldsymbol{w}| \quad \Rightarrow \quad \boldsymbol{u} \times \boldsymbol{u} = \boldsymbol{0}$$

$$\boldsymbol{u} \times \boldsymbol{v} = det\begin{bmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{bmatrix} = w_x\hat{\boldsymbol{x}} + w_y\hat{\boldsymbol{y}} + w_z\hat{\boldsymbol{z}}$$

$$w_x = u_y v_z - u_z v_y$$
$$w_y = u_z v_x - u_x v_z$$
$$w_z = u_x v_y - u_y v_x$$

REVISITING NEWTON'S LAWS: TIME

An example

Particle on a circular orbit: A particle is moving in the x-y-plane on a circular trajectory with the origin as its centre and radius R. It is moving in a counter-clockwise direction with angular frequency ω. Let the particle start from the position $(x, y) = (R, 0)$ at $t = 0$.

1. Write **r**(t).
2. Calculate velocity.
3. Calculate acceleration.
4. What's the direction of velocity and acceleration?

REVISITING NEWTON'S LAWS: TIME

An example

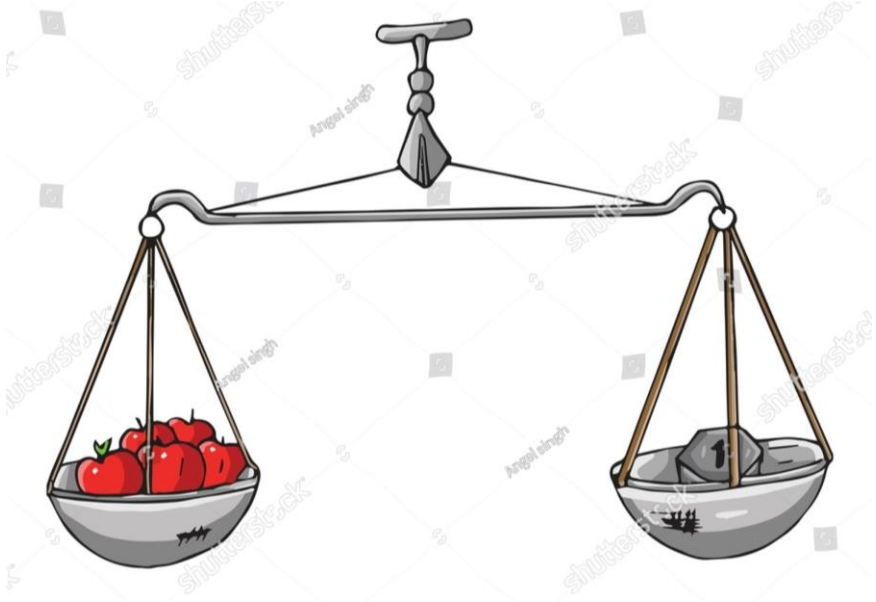
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1. Write **r**(t).
2. Calculate velocity.
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4. What's the direction of velocity and acceleration?

$$x(t) = Rcos(\omega t)$$
$$y(t) = Rsin(\omega t)$$

REVISITING NEWTON'S LAWS: MASS

*The **mass** of an object characterizes the objects inertia – its resistance to being accelerated*



Unit: kg

$$F = mg$$

Weight



REVISITING NEWTON'S LAWS: FORCE

*The **force**, informally, can be considered as a push or pull.*



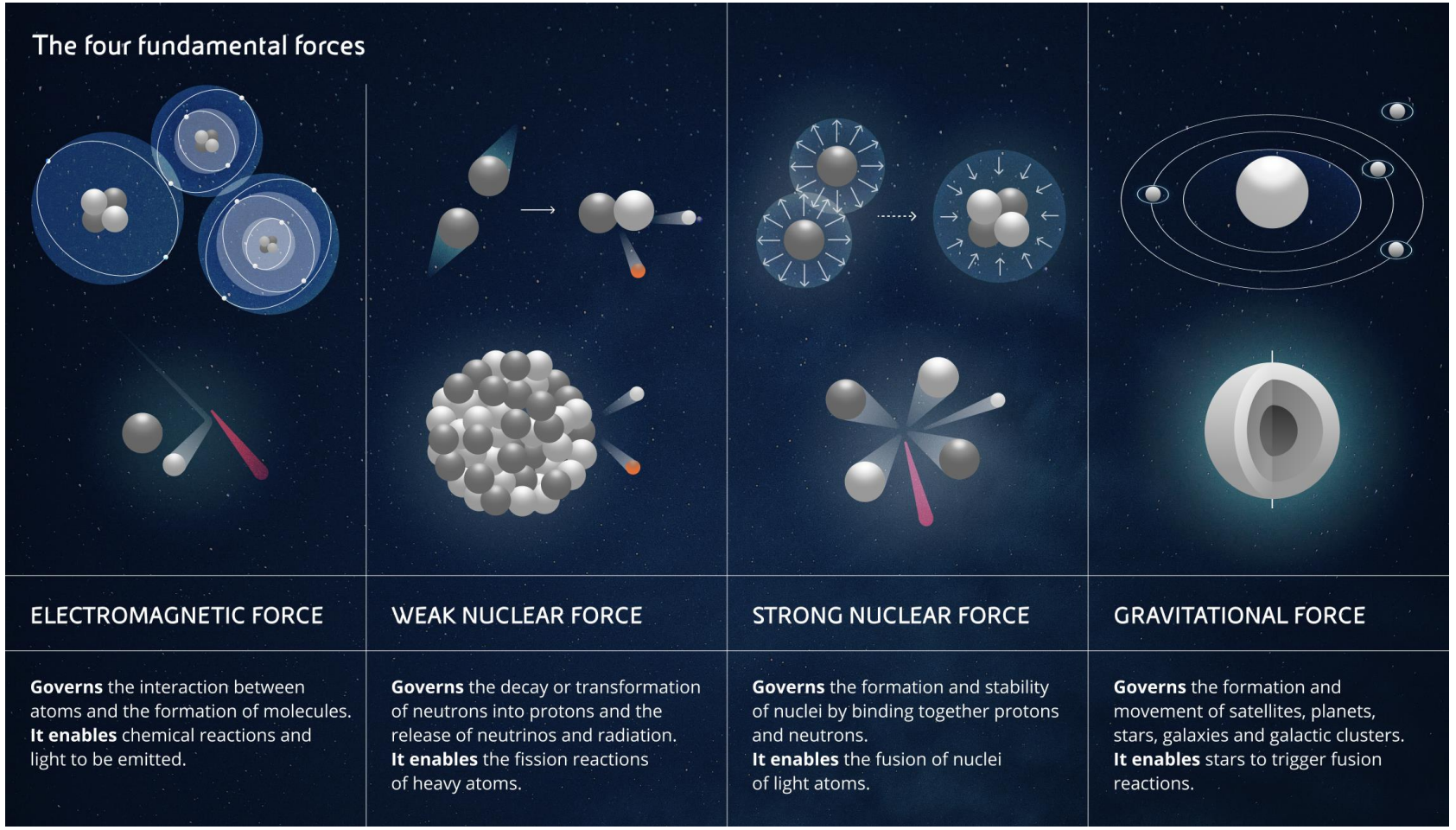
$$F = mg$$

Weight

Unit: N



NEWTON'S LAWS: FUNDAMENTAL FORCES



Forces	Range	Relative strength
Gravitational Force	∞	10^{-38}
Electromagnetic Force	∞	10^{-2}
Strong nuclear force	$< 10^{-15}$m	1
Weak nuclear Force	$< 10^{-18}$m	10^{-13}

NEWTON'S LAWS: DERIVED FORCES



Frictional force

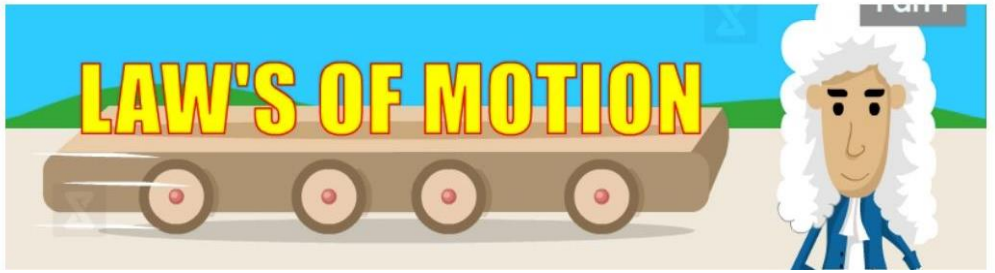


Spring force



Muscular force

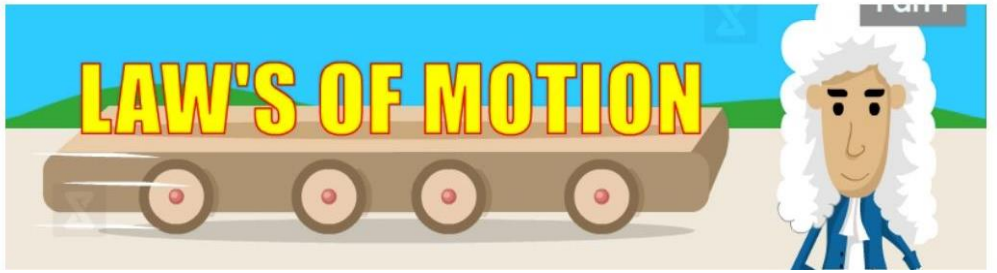
NEWTON'S LAWS:

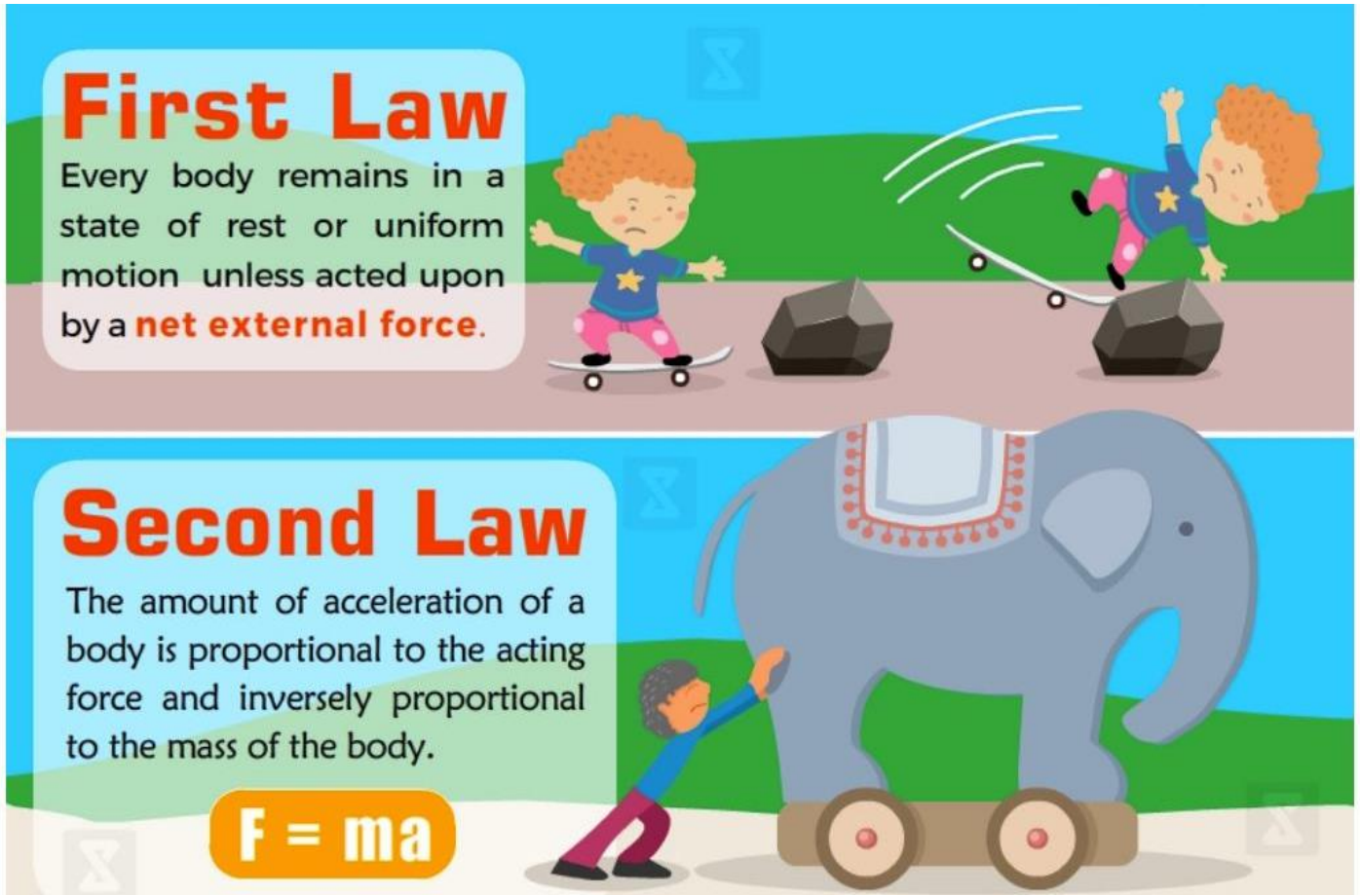






NEWTON'S LAWS:





The concept of a point mass!

A point mass, or particle, is a convenient fiction, an object with mass, with no size, that can move through space but has no internal degrees of freedom.

NEWTON'S LAWS:

$$\boldsymbol{F} = m\boldsymbol{a}$$

NEWTON'S LAWS:

$$\boldsymbol{F} = m\boldsymbol{a} \Rightarrow \boldsymbol{F} = m\ddot{\boldsymbol{r}}$$

$$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \dot{\boldsymbol{v}} = \frac{d^2\boldsymbol{r}}{dt^2} = \ddot{\boldsymbol{r}}$$

Acceleration

NEWTON'S LAWS:

$$\boldsymbol{F} = m\boldsymbol{a} \Rightarrow \boldsymbol{F} = m\ddot{\boldsymbol{r}}$$

$$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \dot{\boldsymbol{v}} = \frac{d^2\boldsymbol{r}}{dt^2} = \ddot{\boldsymbol{r}}$$

Acceleration

Momentum

$$\boldsymbol{p} = m\boldsymbol{v}$$

$$\dot{\boldsymbol{p}} = m\dot{\boldsymbol{v}}$$

NEWTON'S LAWS:

$$\boldsymbol{F = ma} \Rightarrow \boldsymbol{F = m\ddot{r}}$$

$$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \dot{\boldsymbol{v}} = \frac{d^2\boldsymbol{r}}{dt^2} = \ddot{\boldsymbol{r}}$$

Acceleration

Momentum

$$\boldsymbol{p} = m\boldsymbol{v}$$

$$\dot{\boldsymbol{p}} = m\dot{\boldsymbol{v}}$$

$$\boldsymbol{F} = \dot{\boldsymbol{p}}$$

Rate of change of momentum

NEWTON'S LAWS:

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Acceleration

Momentum

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Rate of change of momentum

1D differential equation
(particle confined to move along the *x* axis)

$$\ddot{x}(t) = \frac{F_0}{m}$$

NEWTON'S LAWS:

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Acceleration

Momentum

$$\boldsymbol{p} = m\boldsymbol{v}$$

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Rate of change of momentum

1D differential equation
(particle confined to move along the *x* axis)

$$\ddot{x}(t) = \frac{F_0}{m}$$

$$\dot{x}(t) = \int \ddot{x}(t)\, dt = v_0 + \frac{F_0}{m} t$$

NEWTON'S LAWS:

$$\boldsymbol{F} = m\boldsymbol{a} \Rightarrow \boldsymbol{F} = m\ddot{\boldsymbol{r}}$$

$$\boldsymbol{a} = \frac{d\boldsymbol{v}}{dt} = \dot{\boldsymbol{v}} = \frac{d^2\boldsymbol{r}}{dt^2} = \ddot{\boldsymbol{r}}$$

Acceleration

Momentum

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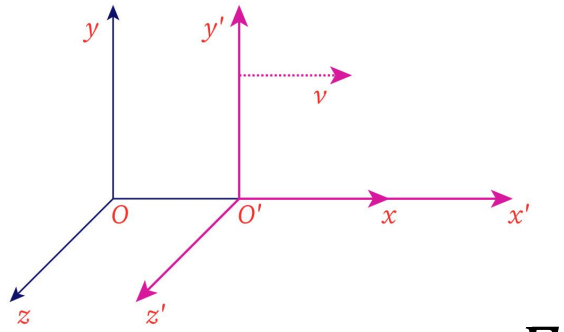
$$\dot{\boldsymbol{p}} = m\dot{\boldsymbol{v}}$$

$$\boldsymbol{F} = \dot{\boldsymbol{p}}$$

Rate of change of momentum

1D differential equation
(particle confined to move along the *x* axis)

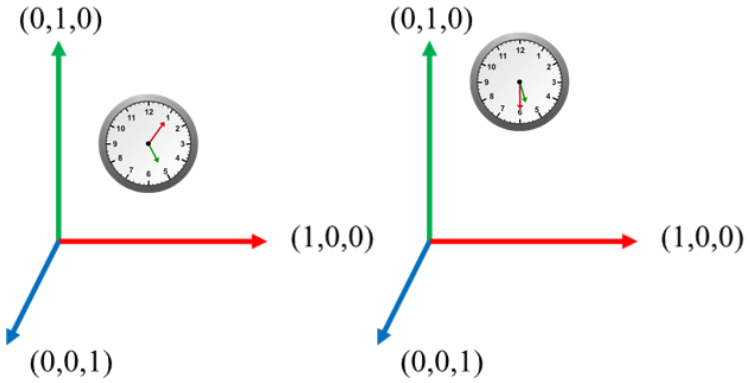
$$\ddot{x}(t) = \frac{F_0}{m}$$



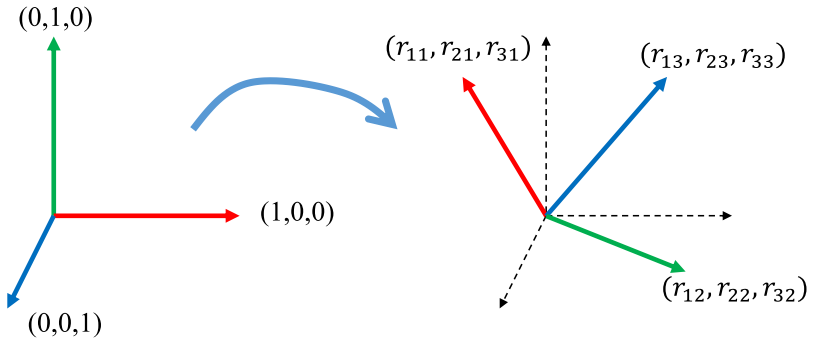
$$\dot{x}(t) = \int \ddot{x}(t)\, dt = v_0 + \frac{F_0}{m}t \Rightarrow x(t) = \int \dot{x}(t)dt = x_0 + v_0 t + \frac{F_0}{2m}t^2$$

x_0, v_0 Initial Conditions

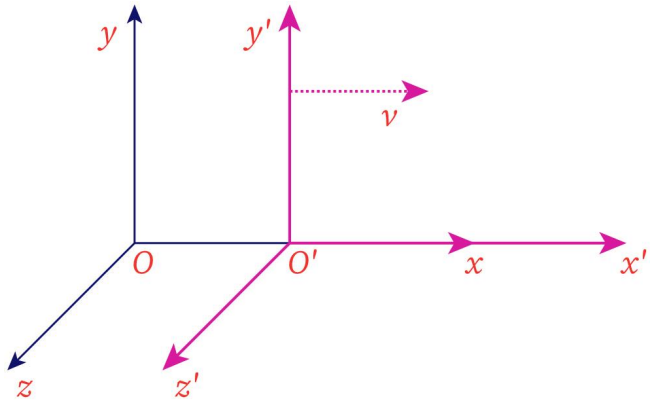
NEWTON'S LAWS: REFERENCE FRAMES



Different time origin



Transformed axes



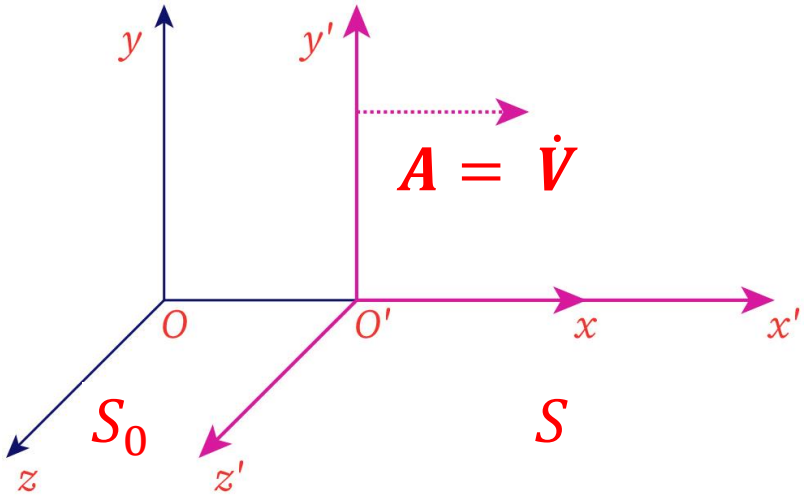
Relative motion

Constant velocity or accelerated

NEWTON'S LAWS: REFERENCE FRAMES

Inertial Frame of Reference	Non-Inertial Frame of reference
A frame of reference which is at a constant velocity with respect to other frames of reference is called Inertial frame of reference.	A frame of reference which is at a constant acceleration with respect to other frame of reference is called non inertial or accelerated frame of reference.
Acceleration of the frame is zero.	Acceleration of the frame is nonzero.
Inertial frame of reference obeys Newton's Laws of motion.	Non-Inertial frame of reference does not obey Newton's Laws of motion.
Example: A train moving with uniform velocity	Example: A freely falling lift

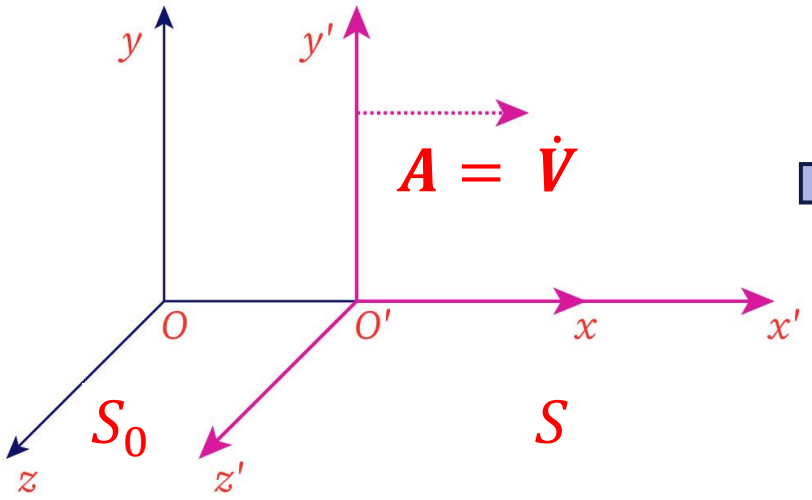
NEWTON'S LAWS: NON-INERTIAL FRAMES



Inertial Frame

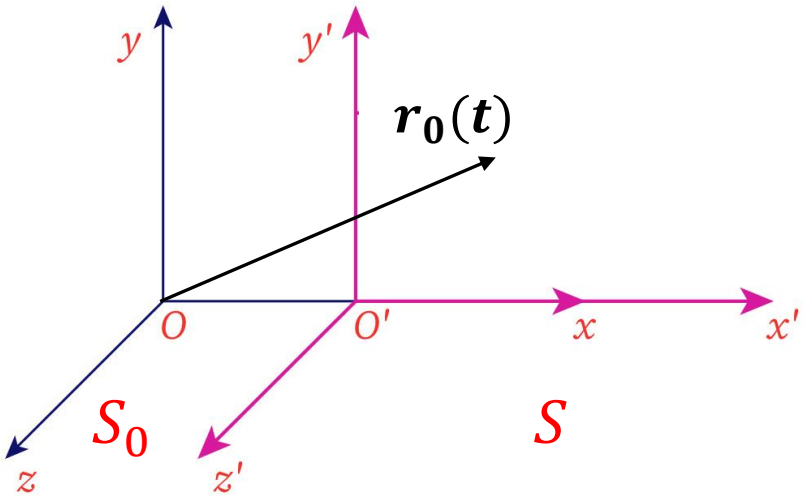
Non-inertial Frame

NEWTON'S LAWS: NON-INERTIAL FRAMES



Inertial Frame

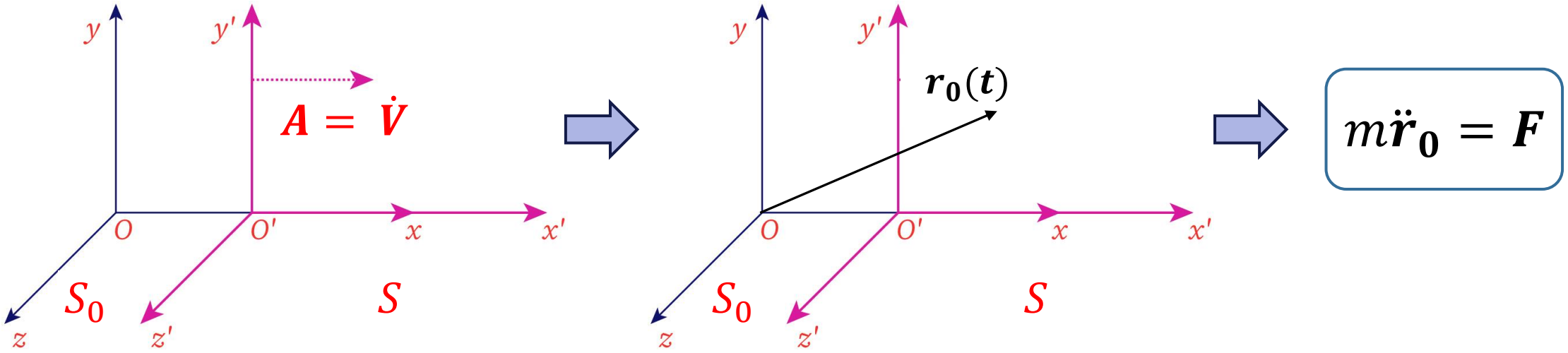
Non-inertial Frame



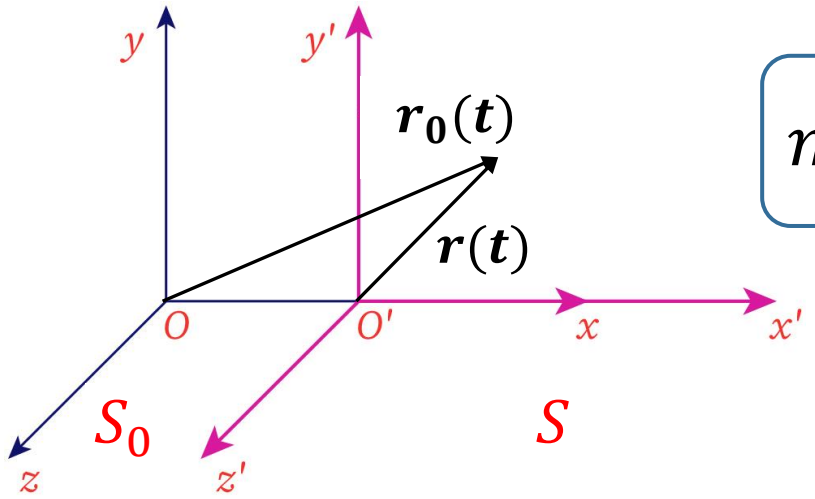
Inertial Frame

Non-inertial Frame

NEWTON'S LAWS: NON-INERTIAL FRAMES



NEWTON'S LAWS: NON-INERTIAL FRAMES

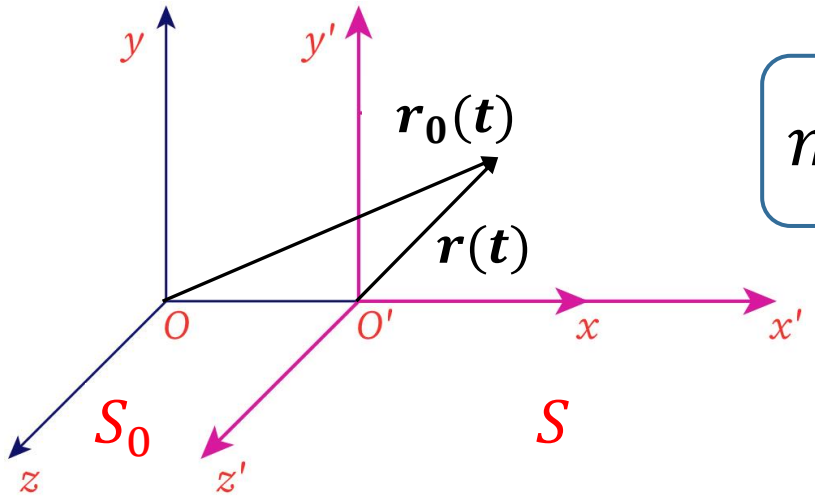


$$m\ddot{\boldsymbol{r}}_0 = \boldsymbol{F}$$

Inertial Frame

Non-inertial Frame

NEWTON'S LAWS: NON-INERTIAL FRAMES



Inertial Frame

Non-inertial Frame

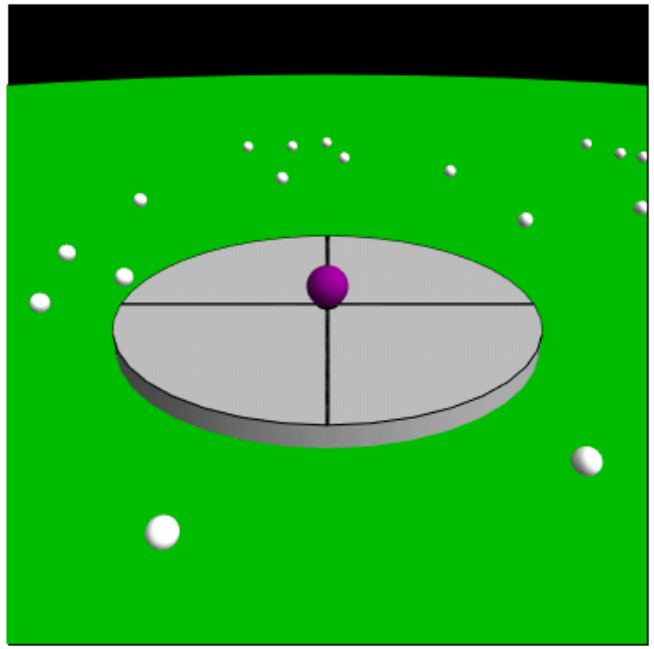
$$m\ddot{r}_0 = F$$

$$A = \dot{V}$$

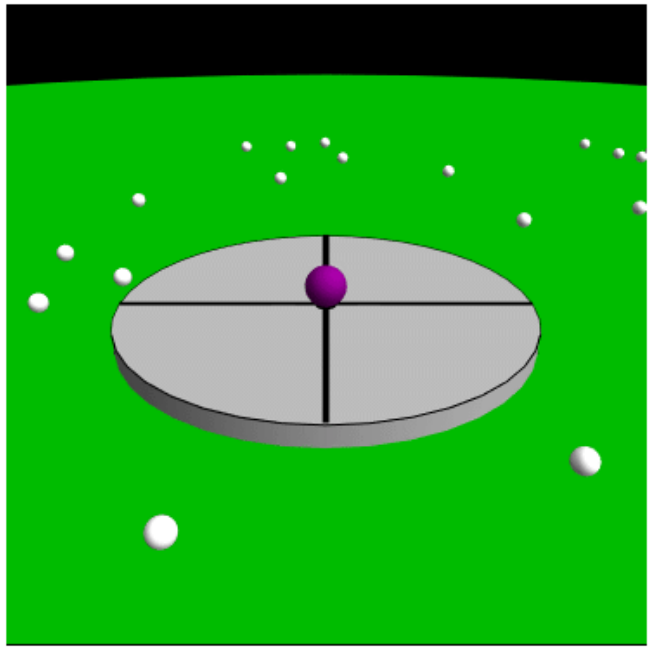
$$m\ddot{r} = F - mA$$

Inertial Force (pseudo force)

NEWTON'S LAWS: NON-INERTIAL FRAMES



Inertial
Frame



Non-inertial
Frame

$$m\ddot{\boldsymbol{r}} = \boldsymbol{F} - m\boldsymbol{A}$$

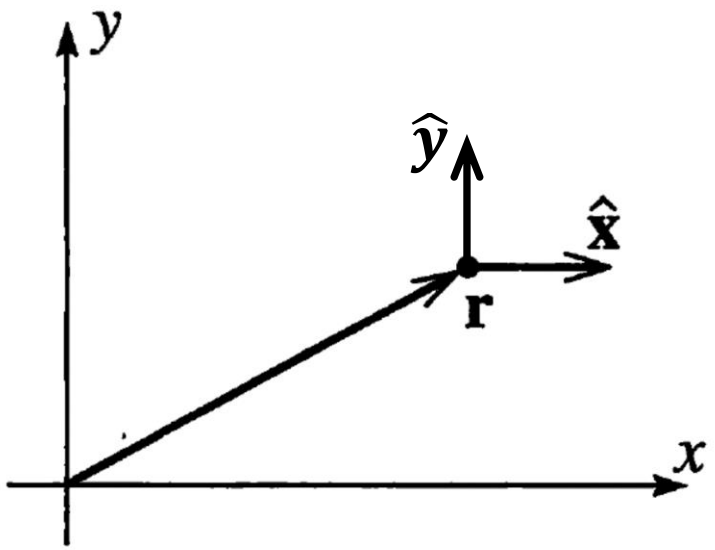
Inertial Force
(pseudo force)

NEWTON'S LAWS: IN 3D (CARTESIAN)

$$\boldsymbol{F} = m\ddot{\boldsymbol{r}}$$

$$\mathbf{F} = F_x\,\hat{\mathbf{x}} + F_y\,\hat{\mathbf{y}} + F_z\,\hat{\mathbf{z}}$$

$$\mathbf{r} = x\,\hat{\mathbf{x}} + y\,\hat{\mathbf{y}} + z\,\hat{\mathbf{z}}.$$



NEWTON'S LAWS: IN 3D (CARTESIAN)

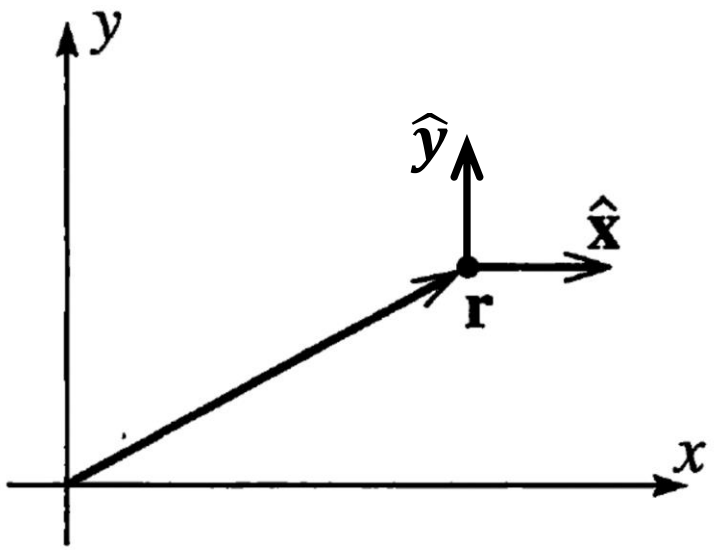
$$\boldsymbol{F} = m\ddot{\boldsymbol{r}}$$

$$\mathbf{F} = F_x\,\hat{\mathbf{x}} + F_y\,\hat{\mathbf{y}} + F_z\,\hat{\mathbf{z}}$$

$$\mathbf{r} = x\,\hat{\mathbf{x}} + y\,\hat{\mathbf{y}} + z\,\hat{\mathbf{z}}.$$

$$F_x\,\hat{\mathbf{x}} + F_y\,\hat{\mathbf{y}} + F_z\,\hat{\mathbf{z}} = m\ddot{x}\,\hat{\mathbf{x}} + m\ddot{y}\,\hat{\mathbf{y}} + m\ddot{z}\,\hat{\mathbf{z}}.$$

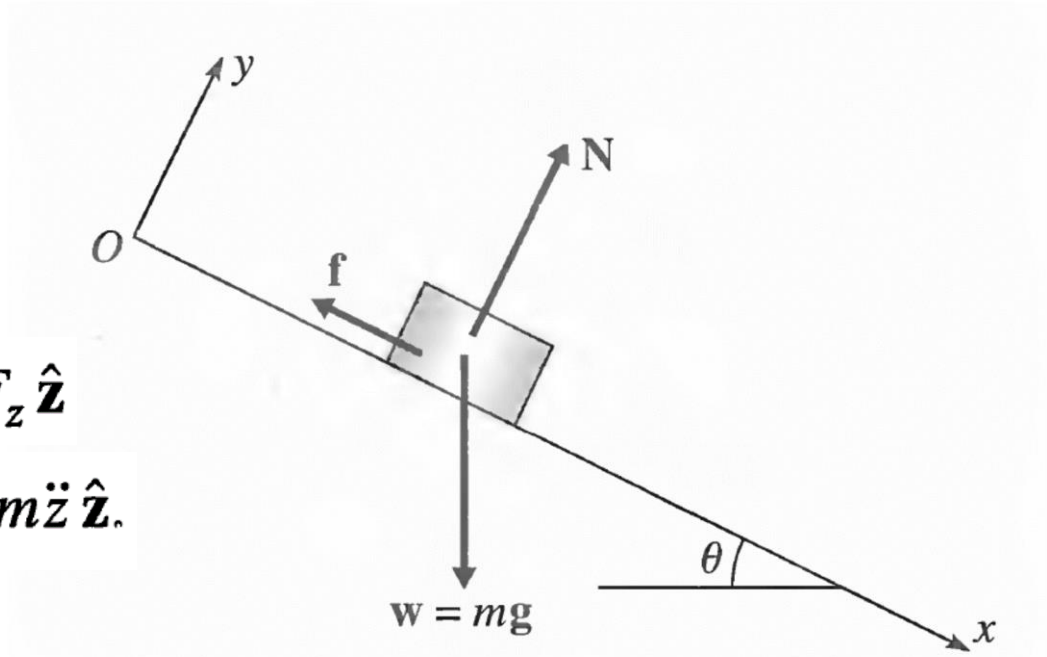
$$F_x = m\ddot{x}$$
$$F_y = m\ddot{y}$$
$$F_z = m\ddot{z}.$$



NEWTON'S LAWS: CARTESIAN COORDINATES

EXAMPLE 1.1 A Block Sliding down an Incline

A block of mass m is observed accelerating from rest down an incline that has coefficient of friction μ and is at angle θ from the horizontal. How far will it travel in time t?



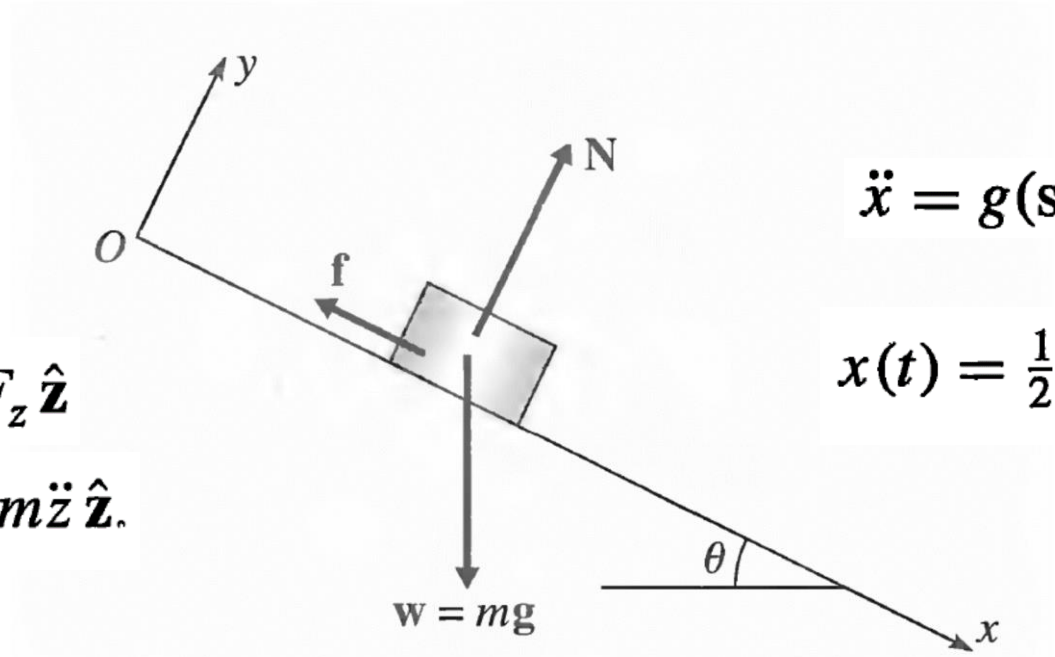
$$F_x\,\hat{\mathbf{x}} + F_y\,\hat{\mathbf{y}} + F_z\,\hat{\mathbf{z}}$$

$$= m\ddot{x}\,\hat{\mathbf{x}} + m\ddot{y}\,\hat{\mathbf{y}} + m\ddot{z}\,\hat{\mathbf{z}}.$$

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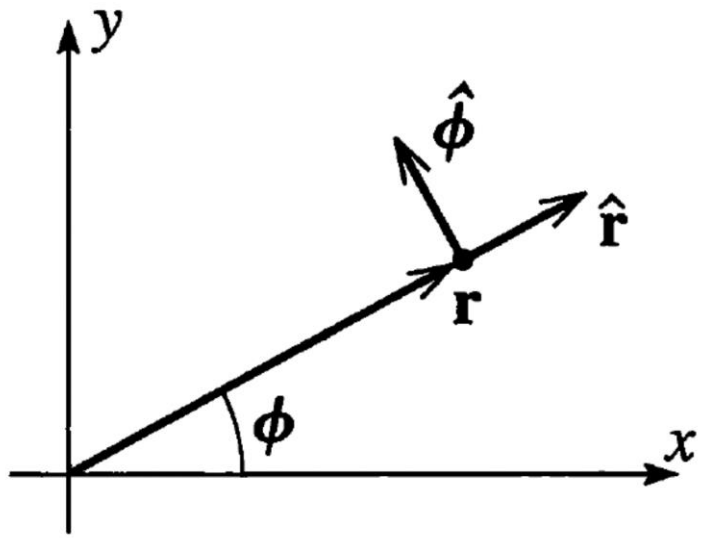
$$= m\ddot{x}\,\hat{\mathbf{x}} + m\ddot{y}\,\hat{\mathbf{y}} + m\ddot{z}\,\hat{\mathbf{z}}.$$

$$\ddot{x} = g(\sin\theta - \mu\cos\theta)$$

$$x(t) = \tfrac{1}{2}g(\sin\theta - \mu\cos\theta)t^2$$

NEWTON'S LAWS: POLAR COORDINATES

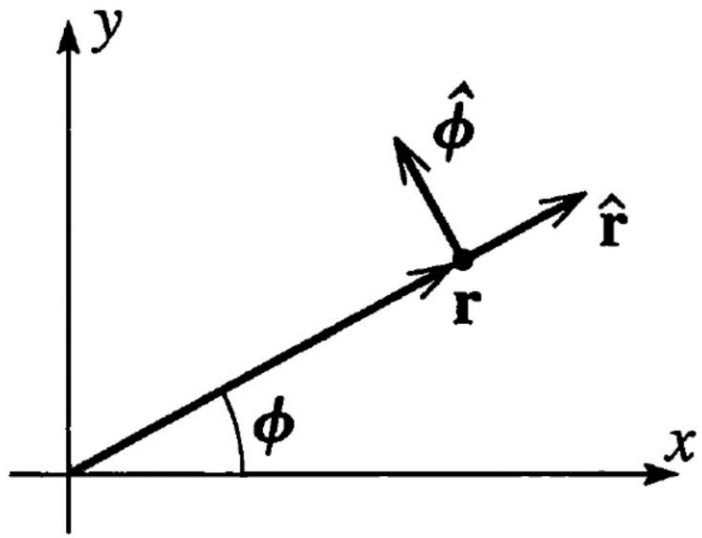
$$\left.\begin{aligned} x &= r\cos\phi \\ y &= r\sin\phi \end{aligned}\right\} \longleftrightarrow \left\{\begin{aligned} r &= \sqrt{x^2+y^2} \\ \phi &= \arctan(y/x) \end{aligned}\right.$$



NEWTON'S LAWS: POLAR COORDINATES

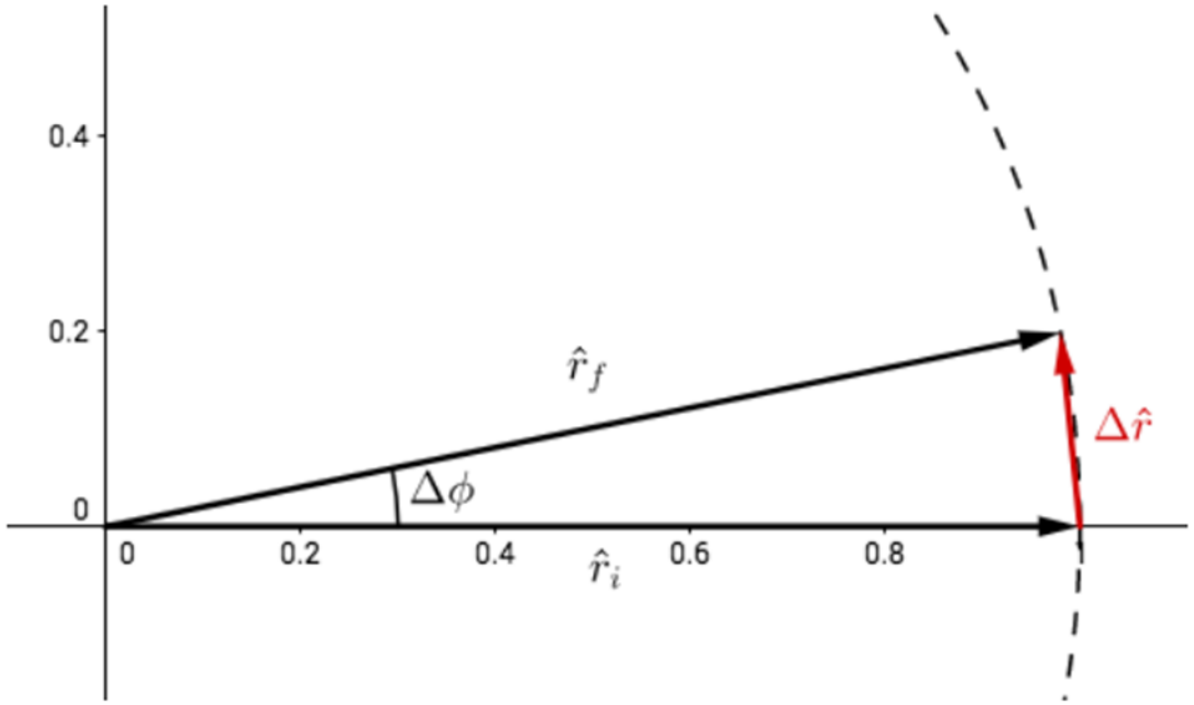
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$$\mathbf{F} = F_r\hat{\mathbf{r}} + F_\phi\hat{\boldsymbol{\phi}}.$$



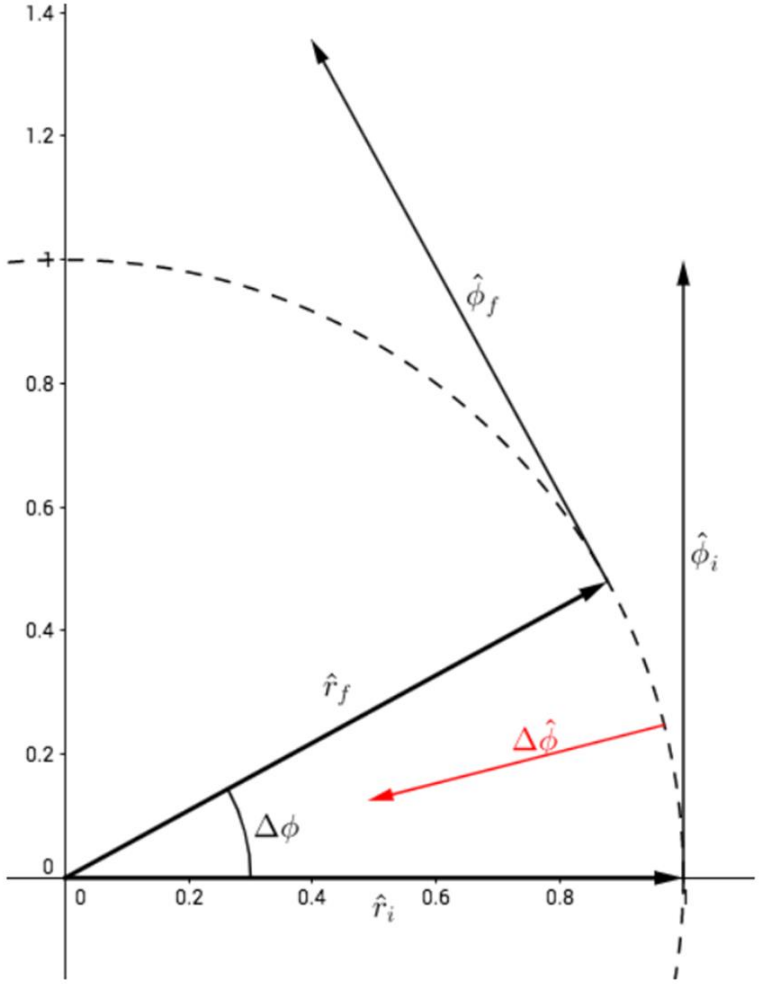
Let's derive expressions for F_r and $F_\emptyset$

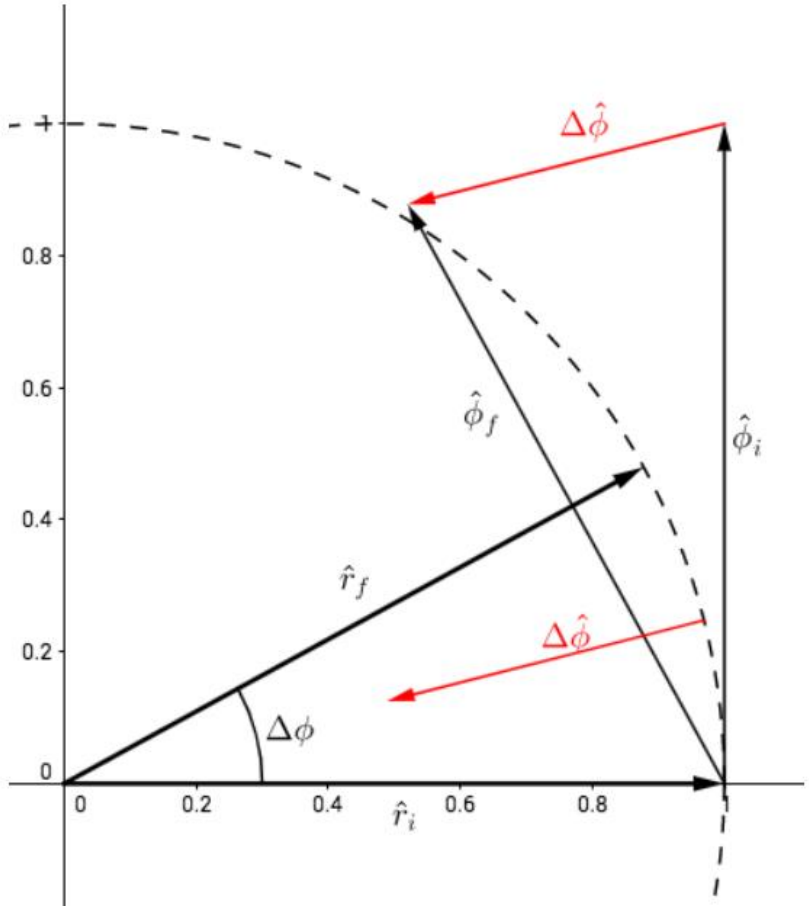
NEWTON'S LAWS: POLAR COORDINATES



$$\Delta\hat{r} = \Delta\phi\hat{\phi}$$

NEWTON'S LAWS: POLAR COORDINATES





$$\Delta\hat{\phi} = -\Delta\phi\hat{r}$$

NEWTON'S LAWS: POLAR COORDINATES

Solution

$$\mathbf{a} = \left(\ddot{r} - r\dot{\phi}^2\right)\hat{\mathbf{r}} + \left(r\ddot{\phi} + 2\dot{r}\dot{\phi}\right)\hat{\boldsymbol{\phi}}$$

$$\mathbf{F} = m\mathbf{a} \quad \Longleftrightarrow \quad \begin{cases} F_r = m(\ddot{r} - r\dot{\phi}^2) \\ F_\phi = m(r\ddot{\phi} + 2\dot{r}\dot{\phi}) \end{cases}$$

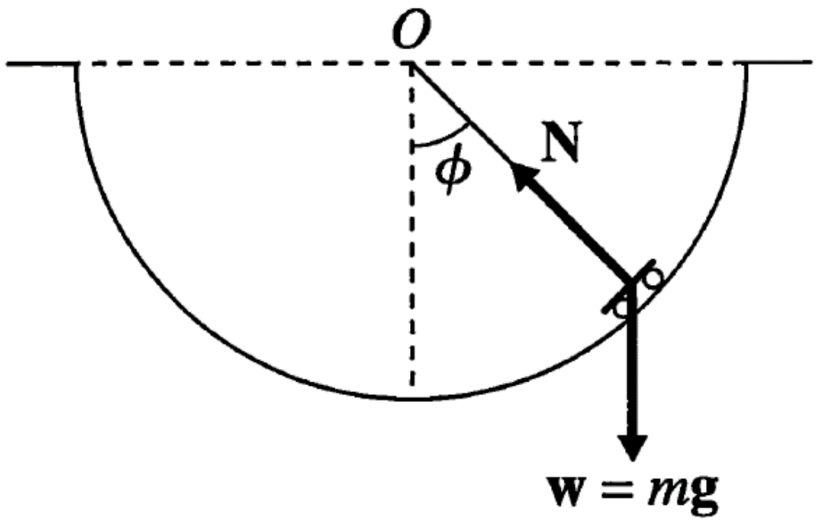
For constant r

$$\mathbf{a} = -r\dot{\phi}^2\hat{\mathbf{r}} + r\ddot{\phi}\hat{\boldsymbol{\phi}} \qquad \mathbf{a} = -r\omega^2\hat{\mathbf{r}} + r\alpha\hat{\boldsymbol{\phi}},$$

NEWTON'S LAWS: POLAR COORDINATES

EXAMPLE 1.2 An Oscillating Skateboard

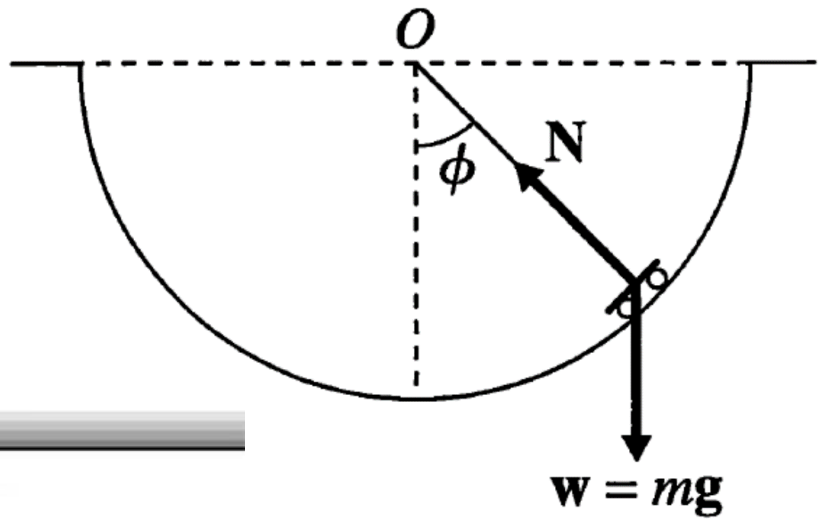
A "half-pipe" at a skateboard park consists of a concrete trough with a semicircular cross section of radius $R = 5$ m, as shown in Figure 1.14. I hold a frictionless skateboard on the side of the trough pointing down toward the bottom and release it. Discuss the subsequent motion using Newton's second law. In particular, if I release the board just a short way from the bottom, how long will it take to come back to the point of release?

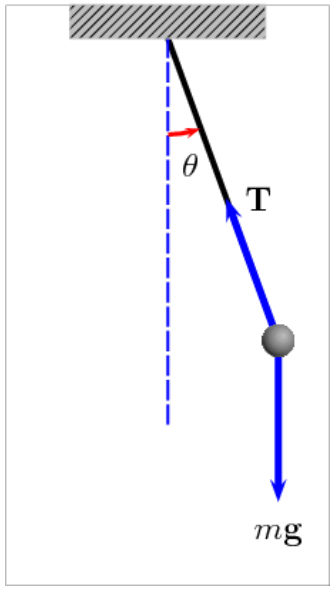


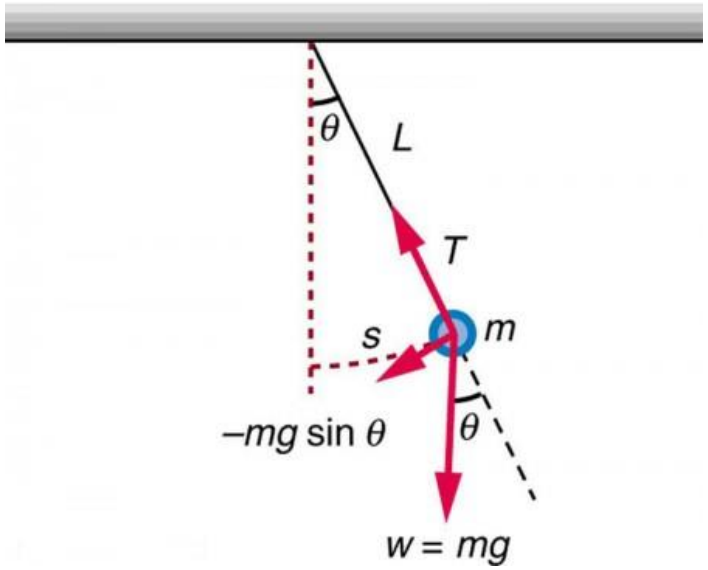
NEWTON'S LAWS: POLAR COORDINATES

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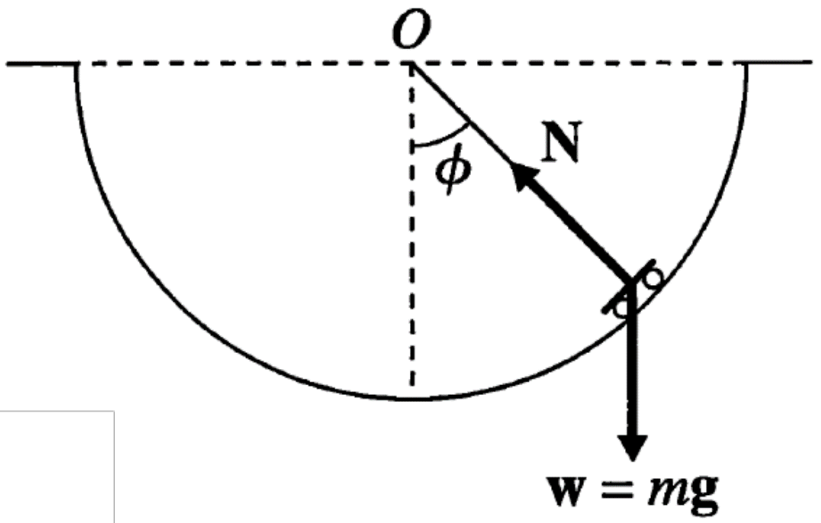




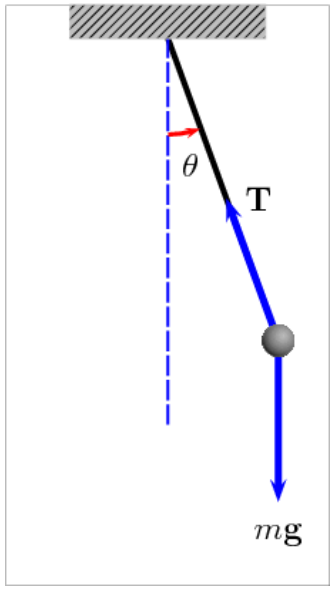
NEWTON'S LAWS: POLAR COORDINATES

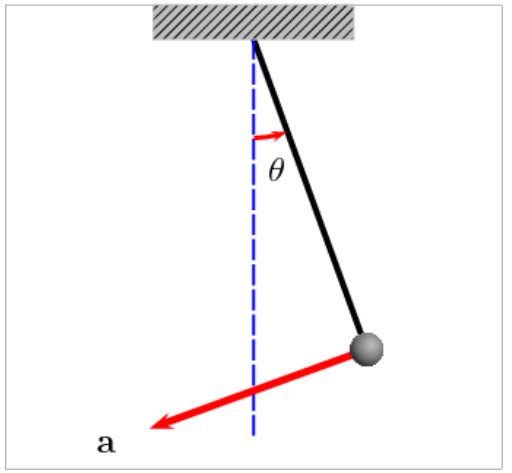
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$$\ddot{\phi} = -\frac{g}{R}\sin\phi.$$

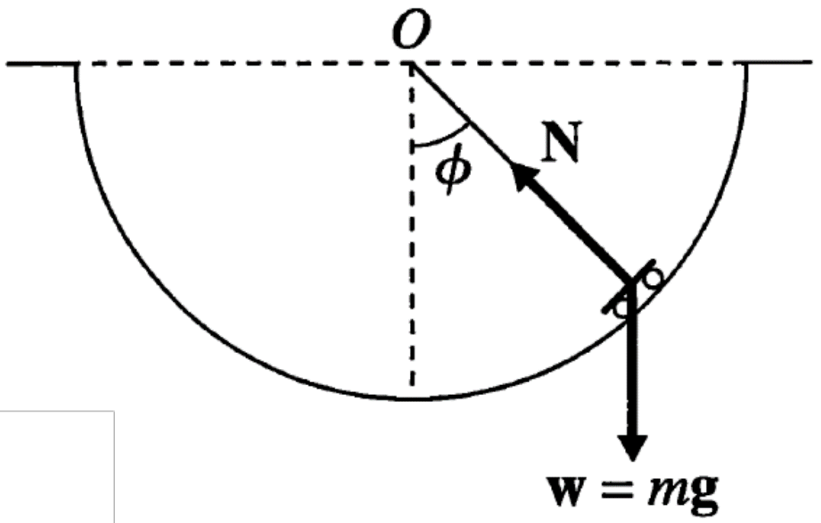




NEWTON'S LAWS: POLAR COORDINATES

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Small angle approximation

$$\omega = \sqrt{\frac{g}{R}}$$

Equation of motion

$$\ddot{\phi} = -\omega^2\phi$$

